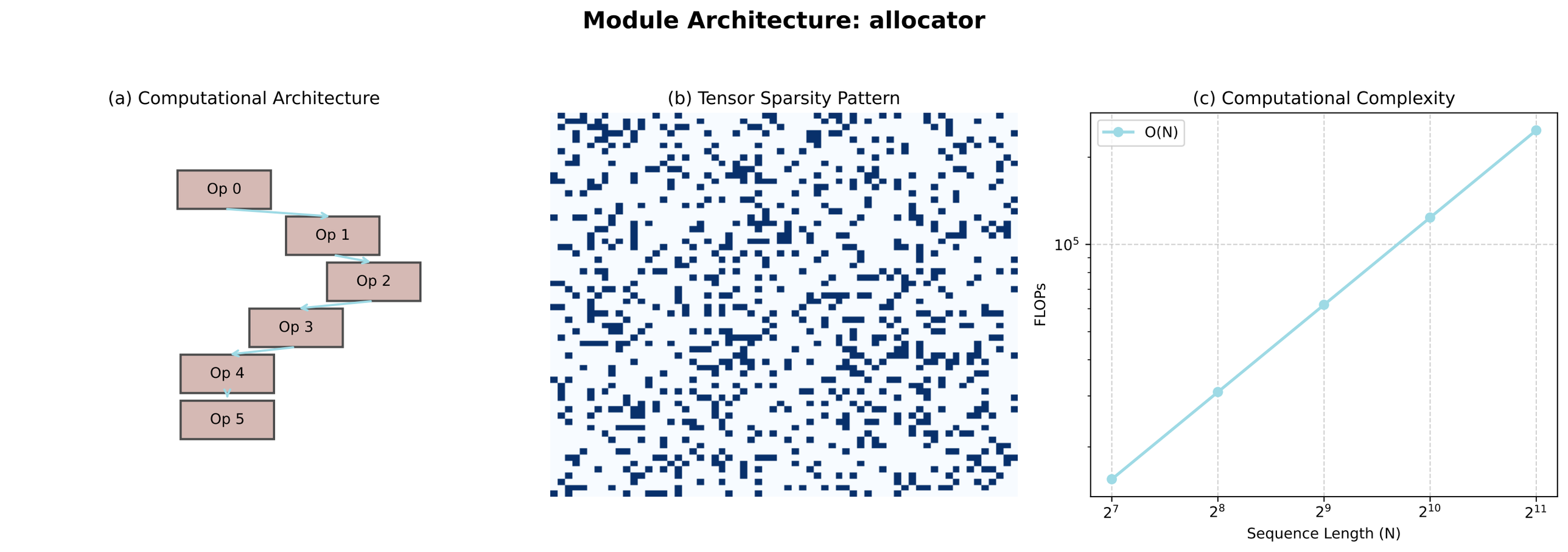Module Architecture: allocator